

DIPOLE TYPE BY LENGTH

l = antenna rod length (m)

λ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m)		
<i>l</i> / λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\ \Omega$
= 1	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199\ \Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors <i>small l</i> ≪ λ $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin\theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$ Power density formulas <i>l</i> / λ < 1/50; <i>far field</i>	
Power density	$S(R, \theta) = S_{\text{max}} \sin^2 \theta$ (W/m ²)
Max pwr density	$S_{\text{max}} = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}$ (W/m ²)
<i>l</i> : rod length (m) <i>R</i> : obs. distance (m)	<i>I</i> ₀ : peak current (A) θ : angle from dipole axis (rad)
$k=2\pi/\lambda$ (rad/m)	$\eta_0=120\pi \approx 377\ \Omega$
Max @ broadside $\perp$ to dipole axis = equator plane $\theta_{\text{max}}=\pi/2$ <i>D exact</i> $D=1.5$ (1.76 dB)	
Total radiated power $P_{\text{rad}}=\frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2$ (W)	

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at <i>R</i> any <i>l</i> $S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$	
<i>At $\theta=0, \pi$: bracket is 0/0. L'Hôpital ⇒ S=0 (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta=0.001$ rad → tiny → 0.</i>	
Normalized pattern dimensionless $F(\theta)=S(\theta)/S_{\text{max}}$	

COMMON ANGLES (deg ↔ rad, trig values)

Conversion multiply by $\pi/180$ or 180/π $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$	
°	rad
sin θ	cos θ
sin ² θ	cos ² θ
0	0
30	$\pi/6$
45	$\pi/4$
60	$\pi/3$
90	$\pi/2$
180	π

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

<i>N</i> identical elements along axis, spacing <i>d</i> , equal phase (broadside, $\theta_{\text{max}} = \pi/2$). Total pattern $S = S_e \cdot F_a$ (S_e = single elem).	
Array factor (uniform, broadside) $k = 2\pi/\lambda$, $\gamma = kd \cos \theta$ $F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\text{max}}=N^2$ at $\theta=\pi/2$	
Normalize $F_a^{\text{norm}}=F_a/N^2$ HPBW (broadside, uniform) use <i>Nd</i>, not $(N-1)d$ $\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) = 50.8° · $\frac{\lambda}{Nd}$ <i>Grating lobes appear if $d > \lambda$. Avoid.</i>	

BEAM PARAMETERS — β , Ω_p , *D*

Normalize always start here $F(\theta) = S(\theta)/S_{\text{max}}$ (unitless) Beamwidth β at threshold <i>X</i> Set $F(\theta_X)=X$, solve for θ_X ; $\beta=2\theta_X$ (symmetric)		
Threshold	$X=10^{\text{dB}/10}$	θ_X (Gauss. short-cut)
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any <i>X</i>	$F(\theta_X)=X$
Pattern solid angle Ω_p recognize pattern, look up $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi \quad (\text{sr})$ <i>No ϕ dep.:</i> $\Omega_p=2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$. <i>θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.</i>		
Pattern $F(\theta)$	Ω_p (sr)	
Isotropic ($F=1$)	$4\pi \approx 12.57$	
Hertzian sin ² θ	$8\pi/3 \approx 8.38$	
Half-wave dipole	≈ 7.66	
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_{1/2}^2/\ln 2$	
2-axis HPBW (aperture)	$\beta_{xz} \cdot \beta_{yz}$	
Other (use integral)	numerical	

TI-36X Pro: [2nd] [d/dx] for $\int_{\square}^{\square}$, RAD mode, multiply by 2 π .

Directivity <i>D</i> always $D=4\pi/\Omega_p$; common shortcuts below $D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}}=10 \log_{10} D$	
WORD TRAP: “direction” = θ_{max} (rad); “directivity” = <i>D</i> (unitless). Dipoles → COMMON DIPOLE PARAMS table. Aperture: $D \approx 4A_p/\lambda^2$. 2-axis HPBW: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$. Circular: $D \approx 4\pi/\beta^2$. Gain (if asked) $\xi = \text{rad. eff.}$, $0 < \xi \leq 1$ $G = \xi D$ $G_{\text{dB}}=10 \log_{10} G$ Lossless / ideal antenna ⇒ $G=D$.	

COMMON DIPOLE PARAMETERS

Lossless ($\xi=1$): $G=D$.				
Type	<i>l</i>	<i>D</i>	<i>D</i> _{dB}	<i>R</i> _{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	< $\lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{\text{rad}}=36.6I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.

ANTENNA AREAS (*A*_e, *A*_p)

Effective area antenna's RX cross section; <i>D</i> from BEAM PARAMS or table above $A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)	
Physical cross section wire dipole, diameter <i>d</i>; <i>l</i> see table above $A_p = l d$ (m ² , any dipole) Inverse: <i>D</i> from <i>A</i> _e $D = \frac{4\pi A_e}{\lambda^2}$ (unitless)	
<i>Thin wire:</i> $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).	

FRIIS TRANSMISSION FORMULA

For each antenna's *G*: if given by NAME (“half-wave dipole” etc.) → COMMON DIPOLE PARAMS (*G* = *D*, lossless). Else (dB/linear value given) → use as given (convert dB → linear).
Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda=c/f$
$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{); } P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for *R max range*
$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using *A*_{e recv. area form}
$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.
General form (off-axis) when patterns NOT peak-aligned
$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

*F*_t, *F*_r normalized patterns at the link directions; *F* = 1 when aligned.
Recipe 1. $\lambda=c/f$. 2. Convert any dB gain to linear: $G=10^{G_{\text{dB}}/10}$.
3. Get *G*_t, *G*_r by antenna type:
• named dipole → COMMON DIPOLE PARAMS (*G* = *D* lossless)
• aperture / dish → APERTURE ANTENNAS ($G=D \approx 4\pi A_p/\lambda^2$)
• arbitrary pattern *F*(θ) → BEAM PARAMS ($G = \xi \cdot 4\pi/\Omega_p$)
• given in dB → dB↔LINEAR table
4. Plug into Friis.
Use linear *G*, not dB. Convert FIRST.

dB ↔ LINEAR

Works for any power-like quantity (<i>G</i> , <i>D</i> , <i>P</i> _r / <i>P</i> _t , SNR, ...): $X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$			
dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB ⇔ ×2, 10 dB ⇔ ×10. Add dB = multiply linear.
For *E*-field / amplitude use 20 log not 10 log (power ∝ *E*²).

NOISE & SNR

Noise power thermal noise into matched receiver $P_N = k_B T_{\text{sys}} B$ (W)	
$k_B = 1.38 \times 10^{-23}$ J/K; <i>T</i> _{sys} system noise temp (K); <i>B</i> receiver bandwidth (Hz). Signal-to-noise ratio $\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}}=10 \log_{10} \frac{P_{\text{rec}}}{P_N}$ Typical comm. link needs SNR > 10 dB for usable signal.	

APERTURE ANTENNAS (horn, dish)

*A*_p physical aperture area (m²); *A*_e effective area (m², = $\lambda^2 D/4\pi$); β_{xz}, β_{yz} HPBW_s in two planes (rad); ℓ , *d* aperture side / diameter (m).
Directivity (any shape, uniform illum.)
 $D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) ⇒ $A_e \approx A_p$
Directivity from beamwidths any aperture
 $D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}$ (use rad); circular: $D \approx 4\pi/\beta^2$

HPBW and <i>D</i> by aperture shape uniform illum.; β in <i>rad</i>			
Shape	<i>A</i> _p	HPBW (rad)	<i>D</i> (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x \ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi/\beta^2 = 4\pi \ell^2 / \lambda^2$
Circular (diam. <i>d</i>)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi/\beta^2 = \pi d^2 / \lambda^2$

Scaling (any ratio) $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$
$$D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}}\right)^2$$

$$\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$$

If a parameter is unchanged, its ratio = 1. Examples: double area → $D \times 2$, $\beta \times 1/\sqrt{2}$; double freq → $D \times 4$, $\beta \times 1/2$.

DIPOLE

l — antenna length (m)
 $\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s
 I_0 — peak current (A)
 $k = 2\pi/\lambda$ (rad/m)
 R — obs. dist. (m); θ from axis
 $\eta_0 = 120\pi \approx 377 \Omega$
 $S(R, \theta)$ — pwr density (W/m²)

BEAM

$F(\theta) = S/S_{\max}$ (unitless)
 a — coef. on θ^2 in Gaussian
 β — HPBW (rad or °)
 Ω_p — pattern solid angle (sr)
 $D = 4\pi/\Omega_p$ — directivity
 ξ rad. eff.; $G = \xi D$
 $D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIIS / dB

P_t — power transmitted (W)
 P_r — power received (W)
 G_t — transmit gain (linear)
 G_r — receive gain (linear)
 $S_r = G_t P_t / (4\pi R^2)$ (W/m²)
 $P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
 $G = 10^{G_{\text{dB}}/10}$
 $3 \text{ dB} = \times 2, 10 \text{ dB} = \times 10$
 $P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m²)
 $A_e = \lambda^2 D / (4\pi)$ (m²)
 $A_e \approx A_p$ for aperture
 $D \approx 4\pi A_p / \lambda^2$ (unitless)
 $\beta \approx 0.88\lambda/\ell$ (rect/sq.)
 $\beta \approx 1.02\lambda/d$ (circular)
 $2A_p \Rightarrow 2D, \beta/\sqrt{2}$

ARRAY

N — number of elements
 d — spacing (m)
 $\gamma = kd \cos \theta$
 $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
 $F_a^{\max} = N^2$ at $\theta = \pi/2$
 $F_a^{\text{norm}} = F_a / N^2$
 $\beta \approx 0.88\lambda/(Nd)$ (broadside)